Does PageRank find the smart contracts users rely on sooner than counting? A crawl-ordering test on an Ethereum rollup

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Abstract

An Ethereum rollup holds millions of smart contracts; wallets, explorers and auditors must choose which to examine first. On the web, a crawler ordered by PageRank on the pages read so far reached important pages sooner than one counting backlinks. We repeated that experiment on Arbitrum One. Nodes are contracts other than smart accounts; edges are ERC-20 transfers and balance-capped allowances between contracts, on five tokens over twelve months. Importance is user-account demand, which no ordering reads: the dollars user accounts paid a contract and authorized it to spend. This analysis and test were defined after the results of a registered analysis were known; the test decides no rule of the plan or the registration. In the deciding run (1,169,044 contracts, 20 seeds), a crawl ordered by a walk coupling both edge types reached the most-demanded contracts sooner than a breadth-first crawl (difference in discovery-curve area +0.0520 [95% interval +0.0485, +0.0558] for inflow; +0.0061 [+0.0014, +0.0111] for authorized value) but not sooner than a crawl ordered by the number of contract backlinks, which was sooner (−0.0051 [−0.0087, −0.0010]; −0.0571 [−0.0621, −0.0519]). Outside the decision, the transfer walk AWP had the highest area and exceeded its backlink count; walks on allowances did not exceed theirs on inflow; in a sensitivity analysis revealing out-edges only, the coupled walk led both comparators. Every crawl stayed far from the ideal order. Frozen before two three-month windows, EndorseRank (PageRank on allowances) ranked new approval pairs below the count of approving contracts (Kendall $\Delta\tau$ −0.036 [−0.046, −0.027]; −0.034 [−0.043, −0.025]); AWP ranked new senders slightly above the transfer in-degree (+0.012 [0.003, 0.023]; +0.011 [0.001, 0.022]) but below it among first-window contracts another contract had paid (−0.035 [−0.049, −0.022]). The labels count mostly user accounts; about 27% of the ranked contracts score zero.

Introduction

An Ethereum rollup holds very many smart contracts. On Arbitrum One [1], the approvals and transfers of five widely used ERC-20 tokens over the eighteen months that this study reads name 25,537,280 addresses, and 1,829,719 of them are contracts that are not the account of a user. Decentralized finance (DeFi) runs lending, trading and asset management through such contracts [2, 3]. Whoever uses them or builds on them has to choose which contracts to look at first: wallet software that shows a request

to sign, a block explorer that labels addresses, an auditor with limited hours, an integrator that composes other protocols. The question behind these choices is which contracts users rely on.

Trust in a contract forms in a few ways today. Most services begin with an authorization: the owner of a token lets a contract move some of it, as an ERC-20 allowance set by calling `approve` or by signing an EIP-2612 `permit` [4, 5]. Most approvals on Ethereum are for unlimited amounts [6], phishing sites lead users to sign exactly such approvals [7], and the toolkits that do so are sold as a service [8]. Owners decide when their wallet shows the approval request, and flaws in how wallets display and check approvals leave that moment open to fraud [9]. Code analysis judges a contract by its functions, for instance by flagging functions written for a rug pull [10]. Deployed reputation tools such as Human Passport weigh credentials to decide whether an address belongs to a distinct person [11], a question about the holder of a key that has no direct counterpart for a contract. None of these reads the public record of which contracts other contracts pay and authorize.

The web met the same question at a larger scale, and graph ranking was its answer. PageRank gives every page the stationary probability of a damped random walk along links [12, 13]. Cho, Garcia-Molina and Page then asked in which order a crawler should visit the pages it has discovered so as to obtain the important ones first [14]. A crawler that has read some pages knows only their links. They ordered its queue by breadth-first discovery, by the number of backlinks seen so far, or by PageRank computed on the links seen so far, and measured how soon each order reached the pages that were important on the whole collection. On 179,000 pages of one university's web, with every crawl started at the university's home page, the PageRank ordering reached the pages with many backlinks sooner than the backlink ordering did, and the authors called PageRank an excellent ordering metric.

On account-based ledgers the PageRank walks that this study builds on run over payments between all addresses. Do and Do apply PageRank and HodgeRank to Ethereum transactions [15], and the Adaptive Weighted PageRank (AWP) of Do, Do and Nguyen weights each transfer by its value and age and restarts the walk at addresses that sent recently [16]. Most addresses of a ledger are accounts of users, not contracts: of the addresses typed in this study 21,877,279 are user accounts, and fewer than half of the event rows, 603,475,784 of 1,240,275,042, have a contract at both ends. Creating an account costs nothing, so one entity can present many identities [17]: wash trading on two decentralized exchanges ran mostly through self-trades and structures of two accounts [18], and airdrop hunters meet eligibility criteria with many identities [19]. A walk over all addresses is therefore moved by edges that are cheap to make. This study walks a graph whose nodes are contracts only. Each node then costs a deployment, and each edge records a relation that one contract holds toward another: a payment, read from a transfer, or an authorization, read from an allowance. An allowance is counted only up to the balance that the authorizing contract holds, because a spender cannot move more than that.

Whether a walk on such a graph finds the contracts that users rely on is a question of evaluation design, and two habits of the field make it hard to answer. One is to check a score against statistics of its own edges. Such agreement is expected by construction: it speaks to the coherence of the intended construct and not to what else the score can tell [20, 21]. Importance therefore has to be measured on something that the orderings do not read, and here it is the demand of user accounts, whose edges enter no graph. The other habit is to compare walks only with other walks. A PageRank smooths a local count, and local counts are hard to beat. New links attach preferentially to nodes that already have many [22, 23], degree is a standard baseline in link prediction [24], also on Ethereum [25], and where degree correlations are weak PageRank is on average

proportional to in-degree [26]. A propagated score earns its extra cost only where it beats the count underneath it.

This paper asks whether PageRank walks on the contract graph of Arbitrum One find the contracts that users rely on sooner, and rank their later use better, than the counts they smooth. Two documents were committed before their data, and we report them with their commits: an analysis plan before this study extracted any log, and a registration of the second label window before this study extracted its logs. The registered analysis that these documents fixed kept amounts in raw token units over every ERC-20 token, scored every event since 1 October 2023 and built graphs that hold every account of the cohort's events. After its results were known, we defined the analysis reported here, the contract-graph analysis: amounts in US dollars on five tokens, scores from the twelve months before each freeze, graphs whose nodes are contracts other than smart accounts with edges read from the whole chain, each allowance counted up to its owner's balance at the freeze, with balances reconstructed from `Transfer` logs, constants set on the contract graph, and a discovery-order test that repeats the crawl-ordering experiment on the contract graph. No result of the contract-graph analysis is a registered result, the test decides no rule of the plan or the registration, and the registered results are given beside the main ones. There are three contributions.

- A discovery-order test on a graph of smart contracts, with importance that no ordering reads. A crawl starts at a seed contract, reads the edges of the contracts it visits and chooses the next contract by an ordering computed on what it has read so far. The decision of the test concerns a walk that couples allowance and transfer edges, C-PR. In the deciding run the crawl ordered by this walk reached the contracts with the most user-account demand sooner than a breadth-first crawl, but not sooner than a crawl ordered by the number of contract backlinks, and the backlink ordering was sooner. Outside the decision, the crawl ordered by AWP had the highest area under the discovery curve and exceeded the transfer backlink count that it smooths on both importances, the walks on allowance edges alone did not exceed the allowance backlink count except in one case, and in the other reveal mode, where a visit shows a contract's out-edges only, the coupled walk led both comparators. Every crawl stayed far from the ideal order.

- An out-of-window comparison of each single-layer walk with the raw count it smooths, on the contract graph, in two windows. Neither the plan nor the registration named these comparisons, and all were computed after the labels were known. EndorseRank, PageRank on the latest allowances, ranked later new approval pairs below the count of approving contracts in both windows, with uniform and with activity restarts. AWP ranked later new transfer senders slightly above the transfer in-degree on the whole spender cohorts and, in the first window, below it among the spenders that another contract had paid before the freeze.

- An exact expression for EndorseRank on spenders whose owner contracts receive no allowance: a common constant times one plus the damped owner-normalized in-strength. It holds within 1% for every such spender, so for most spenders the walk is a reweighted count of approving contracts. With it we report what the walks cost: the time of one solve on each graph, and the solves that a walk-ordered crawl needs and a count-ordered crawl does not.

A companion manuscript by the same authors (unpublished) studies the coupled walk C-PR itself, in the spirit of multilayer ranking [27, 28]: the two decision rules that the plan and the registration fixed for it, applied to the scores of the contract graph as a re-analysis, its weight sweep, its seeded ablation and the Sybil exposure of the allowance

layer. Both papers use the same data, account types and graphs, and both report the spender holdout coefficients of the raw counts, EndorseRank and AWP with their count contrasts, the closed form of EndorseRank and the solver runtimes, with the same values. The discovery-order test is reported only here. The rules for C-PR, its weight sweep, the seeded ablation and the Sybil calibration are reported only in the companion; here C-PR is defined only as far as the test needs it.

Related work

Crawl ordering. The experiment of Cho, Garcia-Molina and Page defines importance metrics (the backlink count, PageRank, similarity to a query and a location metric), ordering metrics computed from the links seen so far, and measures of how soon a crawl obtains the important pages [14]. Its results were not uniform. On a small site of about 1,100 pages the authors report that performance could at times be worse than that of a random crawler, and when the hot pages were those on a given topic, the backlink and PageRank orderings did no better than a random crawler and only the breadth-first crawler gave a reasonable result. Najork and Wiener crawled 328 million pages in breadth-first order and found that this order tends to discover pages of high PageRank early in the crawl; they note that computing PageRank to steer a crawl of that size is very expensive [29]. Baeza-Yates and co-authors simulated crawls of two national web domains with PageRank as the importance of a page and proposed orderings that were more efficient than breadth-first search and than strategies based on partial PageRank calculations [30]. Whether a walk on the pages read so far beats a count therefore depends on the graph, the seeds and the importance measure, and it has to be tested on each kind of graph.

On-chain reputation and risk. Besides the two PageRank studies above [15, 16], RiskProp rates Ethereum accounts by propagating risk through the transaction network [31], and network embeddings of the transaction graph help detect phishing addresses [32]. Surveys of blockchain trust and reputation systems find explicit ratings and transfer activity to be the usual signals [33, 34]. Decentralized credit scoring combines ledger records with prospect theory [35] or machine learning [36], and legal scholarship warns that such scores can become as opaque as the models they would replace [37]. These approaches score accounts by what they send and receive; none orders a crawl of contracts or ranks contracts on the edges between contracts alone.

Contracts as a population. A study of the ERC-20 transfer network finds that externally owned accounts and smart contracts follow different statistical patterns [38], one reason to treat contracts as a population of their own. A measurement of how DeFi protocols are composed of one another, from 23 labeled protocols and their accounts on Ethereum, finds that the accounts of decentralized exchanges and lending protocols have high degree and centrality and that interactions among protocols occur mainly in one strongly connected component [39]. Account labeling infers what an Ethereum account is from public transaction records, for instance with a graph neural network on the transaction network [40]; here an account's type is read from its code and its answers to calls.

Trust propagation and its weak points. HITS separates hubs from authorities [41]. EigenTrust normalizes local trust so that no peer can raise its own standing at will [42], and TrustRank lets trust flow outward from a vetted seed set [43]. Under uniform teleportation that protection is weaker: alliances of link farms lift a target's PageRank [44], and symmetric reputation functions such as PageRank cannot be Sybil-proof [45]. Both results apply to a graph of contracts as well. There each farm address costs a deployment, and under the balance cap its allowance counts only up to the balance it holds; the companion manuscript measures what such a farm gains.

Approvals. EIP-20 specifies the allowance, and EIP-2612 extends it to signed

approvals [4,5]. Wang et al. measure how common unlimited approvals are and how much value they expose [6]. Studies of transaction-based phishing and of drainer services show how attackers obtain approvals [7,8], and Adamczyk dissects an approval-based fraud scheme and the wallet weaknesses it uses [9]. Victor's self-authorization heuristic reads approvals as links between addresses of one entity [46]. An authorization need not express trust: to trade on an exchange a user must first approve its router, and a phishing contract collects approvals as an honest one does. We therefore call the allowance edge an authorization and use the name EndorseRank only as the label of a method.

Baselines and evaluation. Degree is the natural benchmark for predicting new links [24,25], and PageRank often behaves like a monotone function of in-degree [26]. Construct validity separates what a score measures from its agreement with checks built into it [20]. Writing an analysis down before its data are examined limits analytic flexibility [47]; we state for each comparison when it was fixed relative to the data.

Materials and methods

Data, account types and graphs

Events. All events come from Arbitrum One, an optimistic rollup on Ethereum [1], read from the public BigQuery table `goog_blockchain_arbitrum_one_us.logs` (project `bigquery-public-data`). Two queries copied the ERC-20 `Approval` and `Transfer` logs of the whole chain into tables of the authors' project, the first for 1 October 2023 to 30 June 2026 and the second for July to September 2026. Only logs with exactly three topics are read; ERC-721 logs, which share these signatures, carry a fourth topic. The contract-graph analysis reads from these copies the approvals of any value and the non-zero transfers of five tokens from 1 April 2025 to 30 September 2026: 1,240,275,042 rows that name 25,537,280 distinct addresses other than the zero address. The events are those of the whole chain and not those around a cohort. S1 Text gives the topics, decoded fields, label formulas and queries.

Windows. Every score is computed from the twelve months before its freeze, its score window: from 1 July 2025 to 30 June 2026 (the end, 23:59:59 UTC, is written T_{obs}) for the same-window evaluation and the second label window, and from 1 April 2025 to 31 March 2026 (the end is written t_1) for the first. A window of one year describes each contract by its activity in the year before the freeze and gives both freezes windows of equal length. Window W0 freezes the scores at t_1 and counts its labels over April, May and June 2026. Window W1 freezes them at T_{obs}, so its scores are those of the same-window evaluation, and counts its labels over July, August and September 2026. The two score windows overlap from July 2025 to March 2026, and W1's score window contains W0's label window, so the windows are not independent samples.

Tokens and prices. Amounts of different tokens can be added only after they are valued in one unit. We keep the approvals and transfers of five tokens, WETH, USDC, USDT0, ARB and WBTC, and value every amount in US dollars at DefiLlama's daily closing price [48]. We admitted a token if Chainlink's list of data feeds on Arbitrum One shows a market-price feed rated low or medium market risk for the token, for the asset it wraps one to one, or for the Ethereum token of which it is the canonical-bridge copy [49], and if the token occurs in the data. Of the 133 tokens that the list maps to Arbitrum One addresses, 64 qualified and 46 of these have a daily price; the five kept are those with the most rows among the events around the plan's cohort up to T_{obs}, together 75.75% of the rows of all tokens there. Every day of the score and label windows has a price for each of the five tokens. Checked against Chainlink answers read on the chain for 20 of these days per token, the DefiLlama prices were off by a median

of 0.010% to 0.056% per token, with a largest gap of 0.24%. A transfer is valued at the price of its day and a latest allowance at the price of the freeze's day, because the edge represents the authorization that holds at the freeze. Keeping five tokens with reliable prices also keeps out the logs that any token contract can emit in the name of accounts that took no part. Native ether moves without `Transfer` logs and enters the graphs only as WETH.

Account types. Every address of the events that can end an edge between contracts is typed as a user account or a contract, and only contracts are nodes. A user account costs no deployment, the study does not measure who controls it, and user accounts would supply most of a graph's edges and restart mass; in a graph of contracts each node costs a deployment. Four rules type an address. First, an address that sent a transaction is a user account: 17,477,779 addresses. Second, every other address that can be an end of an edge between contracts was read with `eth_getCode` at block 513,216,506 (9 October 2026), the block at which the plan's cohort candidates had been read: the 6,230,270 addresses that sent no transaction and have a counterparty that is neither a transaction sender nor the zero address, with the members of the cohorts and two audit samples. An address without code (2,909,509) or with code that is only an EIP-7702 delegation designator [50] (1,165,025) is a user account. Third, a smart account is a user account, since it is a contract by its code and the account of a user by its use. We recognize two kinds: a Safe [51], which answers its threshold and a non-empty list of owners (46,204), and an ERC-4337 account [52], which answers a canonical EntryPoint and whose code carries a `validateUserOp` selector and no paymaster selector (87,157 by the reads, and 192,656 more whose proxy hides that selector, typed by listed code families). For a proxy, recognized from its code as an ERC-1167 clone [53], an ERC-1967 proxy [54] or a Safe proxy, the code examined is that of its implementation. Fourth, any other address with code is a contract: 1,829,719 addresses. In all 21,877,279 addresses are typed user accounts. An address that sent no transaction and has only transaction senders and the zero address as counterparties cannot end an edge between contracts and is not typed. No call was aborted at every attempt, and none of 1,000 audited transaction senders had code. Smart accounts of kinds that these rules miss stay contracts; Limitations and S1 Text give the counts that are known.

Balances and the balance cap. An allowance can exceed its owner's balance of the token or be unlimited, and the spender can move at most the balance. Each latest allowance is therefore counted up to its owner's balance of the token at the freeze, both valued at the freeze day's price:

$$a^\star = \min(\text{allowance},\ \text{balance of the owner}). \tag{1}$$

An allowance whose owner holds no balance is capped to zero and is no edge. The balance of an address is reconstructed as the net flow of the token's `Transfer` logs to the address up to the freeze, a negative value or one below 10^{-6} token counting as zero. Of 1,500 sampled pairs of token and owner, 1,493 (99.5%) had a reconstructed balance equal to `balanceOf` at the block of T_{obs}; the threshold set for the check was 98%.

The contract graph. Every node of every graph is a contract, and an edge is kept only when both ends are contracts. The allowance layer has an edge from an owner contract to a spender contract when the owner's latest allowance to the spender in the score window, on at least one token, is positive after the cap. The transfer layer has an edge from a sender contract to a recipient contract when the sender made a non-zero transfer to the recipient in the score window. The union graph holds the distinct ordered pairs of different contracts with an edge in either layer. Restarts of a walk run over the contracts of its graph, and no score or count reads any edge of a user account.

Cohorts. The analysis plan's candidate spenders, every address that received a non-zero approval from at least three distinct owners in the extracted logs up to T_{obs},

numbered 29,022, and 27,844 of them held contract code at the block above. The matched cohort consists of the contracts among them that at least three distinct owners approved, with a non-zero amount of one of the five tokens, in the score window of T_{obs}, and that the rules above type as contracts: 6,443 contracts, since 40 of the 6,483 addresses that meet the approval rule are smart accounts. The first freeze has a cohort of its own, drawn by the same rule from its score window so that no event after the freeze chooses it: 6,292 contracts (51 of the 6,343 addresses that meet the rule are smart accounts), 5,351 of which are also in the matched cohort. We call the addresses that meet the approval rule at a freeze its cohort list. The cohort rule counts every approving owner, user accounts included. A spender cohort holds, at each freeze, the contracts that hold a positive latest allowance of one of the five tokens, set in the score window, among the approvals that have an address of the freeze's cohort list as owner or spender: 7,864 contracts at t_1 (of 7,962 listed spenders, 97 are user accounts and one is a system address) and 7,915 at T_{obs} (of 7,995; 79 and one). It includes contracts outside the cohort that a cohort address had approved. A member of a cohort that ends no edge of a graph stays in the cohort and scores zero under that graph's scores and counts. S1 Table summarizes events, account types, graphs, cohorts and labels.

Order of the analyses and what was fixed when

The comparisons were fixed at different times relative to the data, so we state the order of events (times in UTC; S1 File gives excerpts of the plan and the registration).

- At 13:41:27 on 9 October 2026 an analysis plan and a versioned configuration were committed to the authors' Git repository (commit `d060f3a` of its public copy; see Data availability); no log of this study had yet been extracted. Together they fixed the windows, the cohort and account-type rules, every score with its damping, decay, value transform and restart rule, EndorseRank's uniform restarts with the activity-restart variant to be reported beside it, the raw degrees, the same-window checks, the W0 labels, the statistics and one decision rule, whose object is the coupled walk. Amounts were to stay in raw token base units over every ERC-20 token, and every score was to use every extracted event from 1 October 2023 up to its freeze. The plan named no comparison of a PageRank with its raw degree.

- The logs from 1 October 2023 to 30 June 2026 were extracted, and the same-window and W0 results of this registered analysis were computed.

- At 16:49:32 the registration of W1 was committed (commit `1747e64`), once the W0 results were in, and before this study extracted any July–September log. It fixed the freeze, the label window, the spender cohort and its two labels for W1 and a second decision rule for the coupled walk. The extraction script refuses to run without the committed registration; it recorded the registration's SHA-256 hash and started at 16:49:33, and it created the query for the July–September logs at 16:49:40, eight seconds after the commit.

- The July–September logs were extracted and the registered evaluation was run. The comparisons of EndorseRank's activity variant with the approval count, and of EndorseRank with AWP in W1, were computed afterwards. The results of the registered analysis, with these later comparisons, were committed in `344017e`.

- After the registered results were known, the revised analysis was defined: amounts in US dollars on five tokens; scores from the twelve months before each freeze; graphs whose nodes are contracts other than smart accounts, with edges read from the whole chain; each allowance counted up to its owner's balance at the

freeze, with balances reconstructed from `Transfer` logs; constants set on the contract graph; and the discovery-order test.

We call the revised analysis the contract-graph analysis. It is the main analysis of this paper, and Results report it first. Its definitions have design reasons. Dollar amounts make the tokens commensurable and let AWP's value weight act; in raw base units that weight was within 10^{-4} of one for almost every amount. A window of twelve months bounds the age of every event behind a score. A graph of contracts admits only nodes that cost a deployment, and an allowance beyond the owner's balance cannot be spent. Because all of this was defined once the registered results were known, no result of the main analysis counts as a registered one, and the Results subsection "The registered analysis" sets the registered values beside the main ones. The discovery-order test was defined after the registered results were known, and it decides no rule of the plan or the registration; its protocol states its own decision rule, given below. Every comparison of a walk with the count it smooths, in both analyses, was computed after the labels of its window were known. The closed form was derived to account for the computed scores and tests nothing. The two rules that the plan and the registration fixed for the coupled walk are reported in the companion manuscript.

Scores

Weighted PageRank

Every score is computed by one weighted PageRank solver. With edge weights $w(u, v)$, out-strength $o(u) = \sum_v w(u, v)$ and a restart distribution $\boldsymbol{\pi}$ over the node set V, the iteration is

$$\mathbf{r}^{(t+1)} = (1 - d)\,\boldsymbol{\pi} + d\,P^{\top}\mathbf{r}^{(t)} + d\Big(\sum_{u:\,o(u)=0} r_u^{(t)}\Big)\boldsymbol{\pi}, \qquad P_{uv} = \frac{w(u,v)}{o(u)} \text{ if } o(u) > 0, \quad (2)$$

where the rows of dangling nodes ($o(u) = 0$) are zero, so that the last term returns their mass through $\boldsymbol{\pi}$. The damping factor is $d = 0.85$; the iteration starts from the uniform vector and stops once the L_1 change falls below 10^{-8} or after 300 iterations. A cohort contract that a graph does not contain scores zero under that graph's methods; the plan fixed keeping it as an isolated node as a sensitivity analysis.

AWP on transfers

Following Do, Do and Nguyen [16], a transfer e worth x_e US dollars, made Δt_e days before the freeze, adds $\sigma(\Delta t_e)\,V(x_e)$ to the edge from its sender to its recipient, with

$$\sigma(\Delta t) = \frac{1}{1 + \exp\big(k(\Delta t - t_0)\big)}, \qquad V(z) = \frac{2}{1 + e^{-bz}} - 1, \qquad b = \frac{\ln 3}{m}, \qquad (3)$$

so that

$$w_T(u, v) = \sum_{e:\,u\to v} \sigma(\Delta t_e)\,V(x_e), \qquad \pi_u \propto X_u = \sum_{v\neq u} \max_{e:\,u\to v} \sigma(\Delta t_e). \qquad (4)$$

The restart weight X_u, which the authors call activeness, sums the time weight of u's latest transfer to each contract it has paid, so a contract that never pays another receives no restart mass. The paper gives no parameter values. The plan fixed $k = 0.01$ and $t_0 = 180$ days. The ages in a score window run from zero to 365 days, and the time weight falls from $\sigma(0) = 0.858$ through $\sigma(90) = 0.711$ and $\sigma(182) = 0.495$ to $\sigma(365) = 0.136$ for an event on the window's first day. The slope is set from the median

amount: $m_T = 160$ dollars, the median transfer between contracts in the W0 score window rounded to two significant digits, so a transfer of the median amount has value weight one half, one of twice that amount 0.8 and one of half of it 0.27.

EndorseRank on latest allowances

EndorseRank puts AWP's weights on the allowance layer. For each token, owner contract and spender contract only the most recent `Approval` in the score window is kept, in block and log order, a latest value of zero deletes the entry, and the capped value $a^\star$ of Eq (1) must be positive. The weight of the edge $u \to v$ is

$$w_A(u,v) = \sum_{\text{token}} \sigma\big(\Delta t^\star(u,v,\text{token})\big)\, V\big(a^\star(u,v,\text{token})\big), \tag{5}$$

where $\Delta t^\star$ is the age of the approval that set the allowance. The slope uses $m_A = 0.071$ dollars, the median positive capped allowance at t_1, unlimited allowances included, since the cap makes each of them finite. The value weight therefore separates capped allowances of a few cents, and from about one dollar upward it equals one to three decimals, so that the time weight alone remains. EndorseRank restarts uniformly over the contracts of the allowance layer. Carrying AWP's rule over to allowances would send the walk back to owner contracts in proportion to how many spenders each has approved, discounted by the age of each latest approval; that rewards the granting of approvals, while EndorseRank is meant to rank the contracts that receive them. The plan therefore fixed uniform restarts and named the variant with activity restarts as a score to report beside EndorseRank. Only approvals inside the score window are read, so an allowance still in force at the freeze but last approved before the window opened has no edge, and an owner that approves an allowance manager such as Uniswap's Permit2 is linked to the manager, not to the contracts the manager authorizes in turn (S1 Text).

Raw counts

The in-approve degree of a contract is the number of distinct owner contracts that hold a positive capped latest allowance for it, and its transfer in-degree is the number of distinct other contracts that have sent it a transfer, both counted at the freeze on the events of the score window:

$$k_A(v) = \big|\{u : a^\star(u,v,\cdot) > 0\}\big|, \qquad k_T(v) = \big|\{u \neq v : u \text{ sent a transfer to } v\}\big|. \tag{6}$$

Each count is the local statistic that one of the two walks smooths, and out of window each serves as the baseline that its walk has to beat. No count includes a user account.

The coupled walk

The discovery-order test also orders crawls by the walk of the companion manuscript, C-PR, which mixes the two layers node by node. Its layers carry dollar amounts and not the bounded weight V: the allowance layer sums $\min(a^\star, U)$ over tokens, with $U = 16{,}000$ dollars, the 99th percentile of single transfers between contracts in the W0 score window, and the transfer layer sums $x_e\,\sigma(\Delta t_e)$ over transfers. With $o_A(u)$ and $o_T(u)$ the out-strengths of u in the two layers and a weight λ on the allowance layer, the row of u is

$$\begin{aligned} P_{uv} &= \alpha_u \frac{w^{\text{allow}}(u,v)}{o_A(u)} + (1-\alpha_u)\frac{w^{\text{transfer}}(u,v)}{o_T(u)}, \\ \alpha_u &= \frac{\lambda\,\mathbf{1}[o_A(u)>0]}{\lambda\,\mathbf{1}[o_A(u)>0] + (1-\lambda)\,\mathbf{1}[o_T(u)>0]}, \end{aligned} \tag{7}$$

so that a contract with out-edges in one layer follows that layer alone, and a contract with out-edges in neither is dangling. Restarts are uniform. The test uses $\lambda = 0.5$ and, as sensitivity, 0.25 and 0.75; $\lambda = 1$ and $\lambda = 0$ walk one layer alone and are written C-PR(λ=1) and C-PR(λ=0). These two are not EndorseRank and AWP: their weights are dollar amounts and their restarts uniform. S-PR walks the transfer layer with restarts drawn from C-PR(λ=1). Since the constants m_T, m_A and U come from the W0 score window, no label window enters them.

The discovery-order test

The test repeats the crawl-ordering experiment of Cho, Garcia-Molina and Page [14] on the contract graph of one freeze. A contract stands for a page and an edge of either layer for a link. A crawl starts at a seed contract; visiting a contract reads its edges with their weights; the frontier holds the contracts that the crawl has discovered and not yet visited; and an ordering, computed only on what has been read so far, chooses the next visit. The question of the test is whether a crawl ordered by C-PR on the contracts read so far reaches the contracts with the most user-account demand sooner than a breadth-first crawl and sooner than a crawl ordered by the number of contract backlinks. The protocol of the test, published with the results (Data availability), states everything below; S1 Text gives further detail.

Universes and reveal modes. A crawl can read a contract in two ways. In the `out` mode a visit reveals the contract's out-edges, as reading a page reveals its links, and every crawl from a seed in the giant strongly connected component (SCC) of the union graph visits exactly the SCC and the contracts reached from it (OUT); the names are those that Broder et al. gave the parts of the web graph [55]. In the `both` mode a visit reveals in-edges and out-edges, as a reader of the chain's logs sees who paid and authorized a contract as well as whom it paid and authorized; the universe is then the weakly connected component (WCC) that holds the giant SCC. We write $\mathcal{U}$ for the universe of a run. The protocol names the primary mode by a rule on the structure of the graph at t_1, read in a pilot that counted the pairs of each layer without weights: if the SCC and OUT together hold fewer than half the contracts of the WCC, the `both` mode is primary and the `out` mode a sensitivity analysis. They held 332,396 of 1,015,867 contracts, so the `both` mode is primary at both freezes. The same pilot raised the protocol's stop flag for a thin allowance layer: 0.54% of the contracts of the SCC and OUT had a capped allowance in-edge, against a threshold of 5%. The first author decided to go on, a decision recorded with the protocol, because the test is the only direct test of its question; what the allowance layer contributes is read from the contrasts reported outside the decision.

Seeds and schedule. Each run uses 20 seeds: the contracts of the giant SCC with the lowest SHA-256 hash of a fixed salt and their address. No contract is named as a seed, because the chain has no counterpart of a home page and a seed chosen by importance would leak the target. After V visits a crawl recomputes its ordering and then visits the $\max(1, \lceil \rho V \rceil)$ highest-scored contracts of the frontier in order, with $\rho = 1/50$; a rule on projected run time set ρ. Scores within 10^{-12} of each other relative to the frontier maximum are tied, and a tie goes to the contract discovered first and then to the lower address.

Orderings. The crawls compared are a breadth-first crawl (BFS), which visits the frontier in order of discovery, and a random crawl; three counts on the visited part, the raw degrees that the walks smooth, namely the number of distinct visited contracts with an allowance edge to the frontier contract (IB'_A), with a transfer edge to it (IB'_T) and with an edge to it in either layer (IB'_U, the number of contract backlinks); and the walks of the previous subsection run on the graph seen so far, with its visited and frontier contracts as nodes: EndorseRank′, EndorseRank-activity′, AWP′, C-PR′ at

$\lambda = 0.5$, 0.25 and 0.75, its layers C-PR′(λ=1) and C-PR′(λ=0), and S-PR′. A prime marks what is computed on the contracts read so far. The solver settings are those above, and each recompute starts from the previous score vector. Three references are not crawls and are computed on the full graph of the universe: the ideal order, which visits the hot contracts first, and the orders by C-PR and by union in-degree of the full graph. They separate the quality of a score from the loss caused by the partial view. A normalized in-strength δ', the depth-one closed form of Results on the mixed weights, is run as a diagnostic outside every comparison.

Importance. Importance is the demand of user accounts, which no ordering sees. For a contract c of the universe, over the score window of the freeze,

$$I_T(c) = \sum_w \min\big(x(w \to c),\, U_{\text{user}}\big), \qquad I_A(c) = \sum_w \min\Big(\sum_{\text{token}} a^\star(w, c, \text{token}),\, U_{\text{user}}\Big), \tag{8}$$

where the sums run over user accounts w, $x(w \to c)$ is the dollar value that w transferred to c and $a^\star$ the capped latest allowance of Eq (1). I_T is the user-account inflow and I_A the user-authorized value. The cap per account, $U_{\text{user}} = 31{,}000$ dollars, is the 99th percentile of the transfers from user accounts to contracts in the W0 score window; it bounds what one large account can add. The two measures see different roles: a contract that users authorize, such as a router, scores high on I_A, and a contract that users pay into, such as a pool, on I_T. Each has its own verdict. Two further importances are defined on the edges that the orderings read, the union in-degree IB_U and C-PR of the full graph (IR), in the manner of Cho et al.; agreement with them is an intended-construct check.

Hot sets and statistic. The hot set $H_q(I)$ holds the contracts of the universe whose importance I is positive and at least the $\lceil q|\mathcal{U}|\rceil$-th largest value, ties included, with $q = 1\%$. Let t_h be the visit at which a crawl from one seed reaches the hot contract h, the seed being visit 1. The statistic is

$$\text{AUC}_x = \frac{1}{|H|} \sum_{h \in H} \overline{\max\Big(0,\, 1 - \frac{t_h}{\lceil x|\mathcal{U}|\rceil}\Big)}, \tag{9}$$

with the bar a mean over the 20 seeds. It equals the normalized area under the discovery curve, the share of the hot contracts visited against the number of contracts visited, up to $x|\mathcal{U}|$ visits; $x = 10\%$ is primary. A random order has $\text{AUC}_{10\%} = 0.05$ in expectation, and the ideal order approaches one. Uncertainty comes from a paired bootstrap over the hot contracts, 400 resamples with seed 42 and one matrix of resample indices shared by every ordering; a contrast between two orderings reports its point value and the 95% percentile interval of its resampled values. The seeds are fixed in this bootstrap, and a bootstrap over the seeds is reported as a sensitivity.

Decision. The decision is read at T_{obs} in the primary mode, on $H_q(I_T)$ and on $H_q(I_A)$, from two contrasts of $\text{AUC}_{10\%}$: D_1, C-PR′ minus BFS, and D_2, C-PR′ minus IB'_U. For each importance the protocol states three verdicts with their wording. V1, both intervals above 0: "Ordered by C-PR on the contracts read so far, a crawl from a random contract of the giant strongly connected component reached the contracts of the universe with the most [user-account inflow | user-authorised value] sooner than a breadth-first crawl and than a crawl ordered by the number of contract backlinks." V2, the interval of D_1 above 0 and that of D_2 not above 0: "… sooner than a breadth-first crawl, but not sooner than a crawl ordered by the number of contract backlinks." V3, the interval of D_1 not above 0: "… did not reach the contracts of the universe with the most [user-account inflow | user-authorised value] sooner than a breadth-first crawl." An interval below 0 adds a sentence naming the ordering that was faster. The answer to the question of the test is the weaker of the two verdicts, so it is yes only when both are V1.

Reported outside the decision. The protocol lists, in this order, what is reported without a verdict: (1) each walk$'$ against the count it smooths; (2) contrasts among the walks, first C-PR$'$ against its transfer layer walked alone and against AWP$'$, which the protocol reads for whether the allowance layer helps discovery; (3) the other values of λ; (4) the hot sets in the manner of Cho et al.; (5) sensitivities, namely $x = 5\%$ and 20%, $q = 5\%$, hot sets without the hubs, importance by the number of user accounts, a bootstrap over the seeds, a schedule that recomputes four times as often, the other reveal mode, and the share of the summed importance held by the contracts visited so far, reported as curves only; (6) the freeze t_1. Two further items of the list, a second curve (the share of the first K visits that lie among the K most important contracts) and the share of the union pairs that a crawl has read, are in the published summaries and are not reported here. Diagnostics record the share of picks decided by a tie and the share of early visits that went to contracts with no user-account demand.

Checks and labels

Same-window checks. Each score is compared on the matched cohort with three families of local statistics computed over the score window of T_{obs} from the events between contracts [16]. The transfer family holds a contract's in-degree and the summed dollar value of its inbound transfers. The allowance family holds its in-approve degree and the summed capped value of the allowances it holds. The Sybil-stability family holds the ratio of distinct inbound counterparties to inbound transfers, the span between its first and last transfer and its number of active months. The first two families come from the same events as the two layers, so a score's agreement with the family of its own edges is an intended-construct check, partly built in, and not evidence of predictive power.

Spender labels. Two counts are taken in each label window, one per edge type. New approval pairs counts the owners that give the spender a positive approval in the window while their pair held no positive latest allowance at the freeze. New transfer senders counts the distinct addresses that send the spender a transfer in the window and had sent it none since the score window opened, self-transfers excluded. Both count events of the five tokens that have an address of the freeze's cohort list at one end, and both count every account, user accounts included: user accounts make up 90.4% of the 748,258 new approval pairs and 86.3% of the 651,474 new senders of W0, and 87.9% of 702,275 and 82.3% of 618,461 in W1. The scores, which read only edges between contracts, are thus asked to rank arrivals that are mostly user accounts. Both labels are sparse: 1,725 of the 7,864 W0 spenders gained any new approval pair over the three months, and 1,416 any new sender.

Statistical analysis

Agreement between a score and a check or label is measured by Kendall's rank correlation [56] in its tie-corrected form τ_b, on raw scores and raw values, so that tied scores stay tied. We write $\tau(a)$ for the coefficient of score a and $\Delta\tau = \tau(a) - \tau(b)$ for a paired contrast. Uncertainty comes from a paired bootstrap over contracts [57] with percentile intervals [58]: the evaluated cohort is resampled with replacement, and one matrix of resample indices is shared by every score, so a contrast keeps the covariance of its two coefficients. The same window and the spender holdouts use 400 resamples with seed 42. With 400 resamples each interval end carries a Monte Carlo error of about 0.13 bootstrap standard errors. Every interval is unadjusted. A contrast is described as a difference when its interval excludes zero. The intervals treat contracts as independent and the graphs, and so the scores, as fixed, although spenders share owners. No p-values are reported, and no causal claim is made. The scores and counts are zero for every

member outside the graph and the counts are integers, so ties are many, and part of a gap between a count and a walk may reflect how τ_b treats ties rather than information.

Closed-form check and runtime

The closed form derived in Results was checked against EndorseRank as computed on the allowance layer at both freezes: for every spender of the checked set the pipeline compares the computed score with the closed form and records whether it lies within 1%. Runtime covers one solver call on prebuilt edge lists, with matrix assembly and power iteration: one untimed warm-up and five timed runs, under identical settings, of one Python implementation (SciPy sparse matrices, NumPy power iteration) on a machine with an Intel Core Ultra 9 185H processor and 64 GB of memory under Windows 11 (Python 3.11.9, NumPy 2.3.2, SciPy 1.16.1). For the crawls we report the number of recomputes, solves and power iterations per crawl, which do not depend on the machine.

Ethics statement

This study used only publicly available blockchain data and public token prices and involved no human participants. The scored addresses are contracts; no user account is scored or reported, user accounts enter only as sums over accounts in the importance measures and the labels, no attempt was made to link any address to a real-world identity, and only aggregate statistics are reported. [ethics clearance reference to be added by the authors]

Data availability

The evaluation code, versioned configuration, SQL queries, analysis plan, W1 registration, the documents that define the contract-graph analysis and the protocol of the discovery-order test, the run manifest and the machine-readable summaries behind every table with their SHA-256 checksums are publicly available from the Git repository `https://github.com/abitocodes/ERC20-Allowance-PageRank-Wallet-Reputation`, folder `experiments`. That repository holds the study's folder of the authors' private Git repository with its commit history: each copied commit keeps the original author, times, message and file contents but has a new identifier, and the repository's file `HISTORY.md` lists the original identifiers, which the extraction manifest and the digest of the registered analysis record. The plan (commit `d060f3a`), the registration (`1747e64`) and the registered results (`344017e`) are in that folder at those commits. The results of the main analysis were committed in `6b389df`: `https://github.com/abitocodes/ERC20-Allowance-PageRank-Wallet-Reputation/tree/6b389df1b5718a876e8fd3e07d975157d32a5773/experiments`. Their digest names the code revision `bb78da3`, and each summary records the revision at which it was computed. The summaries are under `data/3-published-results-for-thesis/` (registered analysis) and its subfolder `revised-contract-12m/` (main analysis). The account types, the graph tables, the per-contract scores and labels and the visit of every hot contract for every seed and ordering are kept as parquet files in a public Google Drive folder, whose link and checksums are listed in the folder's file `drive-manifest.json` (commit `7aa7928`). The script that checks the closed-form values against those files is in `https://github.com/abitocodes/phd-dissertation-draft-en`, folder `analysis/`, and the scripts that generate the tables and Fig 1 of this article are in its folder `manuscripts/A-plos-one/`. Excerpts of the plan and the registration are S1 File, and the queries are in S1 Text. The public BigQuery table is not a versioned snapshot, so

the extracted tables were kept in the authors' BigQuery project, and the repository holds their manifests; the BigQuery jobs of the whole study billed 8.78 TiB and cost USD 54.88. The per-contract files hold addresses only as they appear on the public ledger, with no off-chain identity.

Use of generative AI

Claude (Anthropic), used through the Claude Code agent, was used in this study for more than language editing. Under the first author's direction it wrote and ran parts of the extraction and evaluation code in the first repository. In the second repository it wrote the script of the folder `analysis/`, which copies the closed-form values from the published summaries and checks them against the per-contract files, and the scripts that generate the tables, Fig 1 and S1 Table of this article. It also drafted and revised the text of this manuscript, its supporting information and the cover letter, and checked text and scripts for errors. Every number reported here was checked against the published summaries with scripted checks. The authors take full responsibility for the content of the article.

Results

The contract graph

Table 1 gives the graph at the two freezes. The graph numbers 1,403,959 contracts over both freezes, and the two layers differ by more than an order of magnitude. At T_{obs} the transfer layer holds 2,580,709 pairs of contracts and the allowance layer 115,006; at t_1 they hold 2,269,073 and 17,940. The balance cap accounts for much of the difference. Of the 339,282 positive latest allowances between contracts at T_{obs}, 223,892 (66.0%) are unlimited and 222,559 (65.6%) are capped to zero because the owner contract holds no balance of the token, which leaves 116,723; at t_1, 289,299 of 308,922 (93.6%) are capped to zero and 19,623 remain. The allowance layer at T_{obs} is also concentrated: one owner contract holds 98,073 of its 115,006 pairs, whereas at t_1 the largest owner holds 256. The transfer layer has hubs of its own: its largest recipient ends 441,612 transfer pairs at T_{obs}. The union graph at T_{obs} holds 2,575,632 pairs over 1,299,438 contracts, with a giant strongly connected component of 265,804 contracts, 174,239 contracts that reach it, 242,833 that it reaches and 616,562 others.

Coverage of the cohorts is partial, and it frames every result on them. Of the 6,443 contracts of the matched cohort, 4,270 (66.3%) end a pair of the union graph and 1,250 (19.4%) a pair of the allowance layer; of the spender cohorts, 5,700 of 7,864 (72.5%) in W0 and 5,759 of 7,915 (72.8%) in W1 lie in the union graph. The cohort rule and the spender lists count approvals from every account, so a member can have its three approving owners, or its positive allowance, among user accounts only and end no edge between contracts. Most approvals come from user accounts: of the 3,809,524 pairs of an owner and a W1 spender with a positive latest allowance at the freeze, 3,611,886 have a user account as owner and 197,638 a contract. A member without an edge between contracts scores zero under every score and count. Ties follow: EndorseRank gives 801 distinct scores to the matched cohort and zero to 5,200 of its contracts, and AWP gives 3,982 distinct scores and zero to 2,384.

Discovery order: the decision

In the deciding run, at T_{obs} in the `both` mode, the universe holds 1,169,044 contracts, and the hot sets hold 13,141 contracts by user-account inflow and 11,662 by

Table 1. The contract graph at the two freezes and the fit of the closed form for EndorseRank.

	T_{obs}, 30 Jun 2026 (same window and W1)	t_1, 31 Mar 2026 (W0)
Score window	1 Jul 2025 – 30 Jun 2026	1 Apr 2025 – 31 Mar 2026
Positive latest allowances between contracts[a]	339,282	308,922
unlimited	223,892 (66.0%)	208,245 (67.4%)
capped to zero (the owner holds no balance)	222,559 (65.6%)	289,299 (93.6%)
positive after the balance cap	116,723	19,623
Allowance layer: owner–spender pairs	115,006	17,940
owner contracts / spender contracts	9,459 / 100,112	10,153 / 2,050
pairs of the owner contract with the most pairs	98,073	256
Transfer layer: sender–recipient pairs	2,580,709	2,269,073
Union graph: contracts with a pair / pairs	1,299,438 / 2,575,632	1,111,554 / 2,269,421
giant SCC / IN / OUT / other	265,804 / 174,239 / 242,833 / 616,562	222,592 / 169,237 / 109,804 / 609,921
Cohort of the freeze (contracts)	6,443 (matched cohort)	6,292 (W0 cohort)
in the union graph	4,270 (66.3%)	4,108 (65.3%)
in the allowance layer	1,250 (19.4%)	1,248 (19.8%)
Spender cohort (contracts)	7,915 (W1)	7,864 (W0)
in the union graph	5,759 (72.8%)	5,700 (72.5%)
in the allowance layer	1,975 (25.0%)	1,901 (24.2%)
Closed form of EndorseRank[b]: contracts checked	6,443 (matched cohort)	7,864 (W0 spenders)
spenders among them (allowance in-edge)	926	1,362
depth one (no owner has an in-edge)	651	810
deeper (an owner is itself approved)	275	552
common score c of nodes without in-edges	8.45×10^{-6}	4.59×10^{-5}
within 1% of the closed form: depth-one spenders	100.0%	100.0%
within 1% of the closed form: deeper spenders	33.8%	22.1%
within 1% of the closed form: all spenders	80.3%	68.4%

Every node is a contract other than a smart account, and every edge joins two contracts; the edges are read from the events of WETH, USDC, USDT0, ARB and WBTC on the whole chain inside the score window, the twelve months before the freeze. The graph numbers 1,403,959 contracts over both freezes. The union graph holds the distinct ordered pairs of different contracts with an edge in either layer; the pair counts of the two layers also include the pairs of a contract with itself (15,596 transfer pairs and 25 allowance pairs at T_{obs}; 11,430 transfer pairs and 35 allowance pairs at t_1). The bow-tie (giant strongly connected component, SCC; contracts that reach it, IN; contracts reached from it, OUT) is that of the union graph. A member of a cohort outside a graph scores zero under that graph's scores. [a]The latest `Approval` of each (token, owner contract, spender contract) in the score window, kept when its value is positive; unlimited means at least 1.15×10^{77} base units. Each allowance counts up to the owner's balance of the token at the freeze, so an allowance whose owner holds no balance is no edge. A pair sums its tokens; at T_{obs} 114,997 of the pairs have a positive EndorseRank weight in floating point. [b]Eq (11), derived in Results; a spender is a contract of the checked set with an allowance in-edge of positive weight, and it has depth one when none of its owner contracts has an in-edge itself. Damping $d = 0.85$. Sources: `describe/describe.json` (`allowances`, `pairs`, `nodes`); `discovery/discovery_tobs_both.json` and `discovery_t1_both.json` (`sizes.graph`); `same-window`, `holdout-w0` and `window-w1/eval_summary.json` (`cohort_coverage`, `closed_form`, the last through `analysis/closed-form/closed_form_check.json`); the configuration files for the windows.

user-authorized value (Table 2). The second hot set is every contract of the universe with a positive I_A, since fewer than 1% of its contracts have one. On $H_q(I_T)$ the crawl ordered by C-PR′ has $\text{AUC}_{10\%} = 0.1085$, the breadth-first crawl 0.0565 and the crawl ordered by contract backlinks 0.1136. The first decision contrast is above zero, $D_1 = +0.0520$ $[+0.0485, +0.0558]$, and the second below zero, $D_2 = -0.0051$ $[-0.0087, -0.0010]$. On $H_q(I_A)$ the three crawls have 0.0943, 0.0882 and 0.1514, so that $D_1 = +0.0061$ $[+0.0014, +0.0111]$ and $D_2 = -0.0571$ $[-0.0621, -0.0519]$. The verdict is V2 on both importances, with the added sentence, and the summary records it in the protocol's words: "Ordered by C-PR on the contracts read so far, a crawl from a random contract of the giant strongly connected component reached the contracts of the universe with the most user-account inflow and those with the most user-authorised value sooner than a breadth-first crawl, but not sooner than a crawl ordered by the number of contract backlinks. A crawl ordered by the number of contract backlinks reached them sooner." The answer to the question of the test is therefore no.

Fig 1A and 1B show the discovery curves behind these areas. Every crawl stays far from the ideal order, whose $\text{AUC}_{10\%}$ is 0.9438 on I_T and 0.9501 on I_A against at most

Table 2. Discovery-order test: every ordering in the deciding run, and the decision.

Ordering	$\mathrm{AUC}_{10\%}$ on $H_q(I_T)$ [95% interval]	$\mathrm{AUC}_{10\%}$ on $H_q(I_A)$ [95% interval]
Breadth-first (BFS)	0.0565 [0.0543, 0.0586]	0.0882 [0.0859, 0.0900]
Random	0.0609 [0.0593, 0.0627]	0.0725 [0.0710, 0.0742]
IB'_A (allowance backlinks)	0.0626 [0.0601, 0.0654]	0.0881 [0.0853, 0.0905]
IB'_T (transfer backlinks)	0.1135 [0.1090, 0.1184]	0.1466 [0.1416, 0.1513]
IB'_U (contract backlinks)	0.1136 [0.1091, 0.1184]	0.1514 [0.1462, 0.1562]
EndorseRank′	0.0621 [0.0595, 0.0648]	0.0869 [0.0841, 0.0893]
EndorseRank-activity′	0.0612 [0.0583, 0.0643]	0.1042 [0.1002, 0.1079]
AWP′	0.1465 [0.1411, 0.1524]	0.1582 [0.1521, 0.1644]
C-PR′ ($\lambda = 1$)	0.0617 [0.0593, 0.0645]	0.0864 [0.0836, 0.0888]
C-PR′ ($\lambda = 0$)	0.1061 [0.1012, 0.1114]	0.0806 [0.0760, 0.0850]
C-PR′ ($\lambda = 0.25$)	0.1081 [0.1032, 0.1132]	0.0930 [0.0883, 0.0982]
C-PR′ ($\lambda = 0.5$)	0.1085 [0.1036, 0.1136]	0.0943 [0.0895, 0.0996]
C-PR′ ($\lambda = 0.75$)	0.1091 [0.1039, 0.1141]	0.0952 [0.0905, 0.1004]
S-PR′	0.1069 [0.1019, 0.1121]	0.0861 [0.0817, 0.0908]
δ' (diagnostic, outside every comparison)	0.1643 [0.1587, 0.1708]	0.1942 [0.1874, 0.2012]
References that are not crawls		
Order by C-PR of the full graph	0.1061 [0.1015, 0.1115]	0.1157 [0.1105, 0.1210]
Order by union in-degree of the full graph	0.1593 [0.1534, 0.1663]	0.1994 [0.1933, 0.2055]
Ideal order (hot contracts first)	0.9438 [0.9432, 0.9444]	0.9501 [0.9496, 0.9507]
Random curve	0.0500	0.0500
The decision		
Hot contracts, $\lvert H_q \rvert$	13,141	11,662
D_1: C-PR′ − BFS	+0.0520 [+0.0485, +0.0558]	+0.0061 [+0.0014, +0.0111]
D_2: C-PR′ − IB'_U	−0.0051 [−0.0087, −0.0010]	−0.0571 [−0.0621, −0.0519]
Verdict	V2; IB'_U sooner	V2; IB'_U sooner

Deciding run: freeze T_{obs}, `both` mode (a visit reveals a contract's in-edges and out-edges). The universe $\mathcal{U}$ is the weakly connected component that holds the giant strongly connected component: 1,169,044 contracts and 2,378,418 union pairs. $\mathrm{AUC}_{10\%}$ is the mean over the hot contracts of $\max(0, 1 - t_h/\lceil 0.1|\mathcal{U}| \rceil)$, with t_h the visit at which the crawl reaches hot contract h, averaged over 20 seeds from the giant strongly connected component; a crawl recomputes its ordering each time its visits have grown by the share $\rho = 1/50$. $H_q(I_T)$ and $H_q(I_A)$ hold the contracts of the universe with the largest positive user-account inflow and user-authorized value, $q = 1/100$ of the universe with ties. A primed name is the count or walk computed on the contracts read so far. Brackets: 95% percentile intervals of a paired bootstrap over the hot contracts, 400 resamples (seed 42). V1: both intervals of D_1 and D_2 above 0; V2: the interval of D_1 above 0 and that of D_2 not; V3: the interval of D_1 not above 0; an interval of D_2 below 0 adds that the crawl ordered by contract backlinks was sooner. Statement recorded with the verdict: "Ordered by C-PR on the contracts read so far, a crawl from a random contract of the giant strongly connected component reached the contracts of the universe with the most user-account inflow and those with the most user-authorised value sooner than a breadth-first crawl, but not sooner than a crawl ordered by the number of contract backlinks. A crawl ordered by the number of contract backlinks reached them sooner." The test was defined after the registered results were known; it decides no rule of the analysis plan or the registration. Source: `discovery/discovery_tobs_both.json`, keys `auc`, `decision`, `sizes`.

0.1643 and 0.1942 for any crawl, the diagnostic included. After 10% of the visits the crawl ordered by AWP′ had reached 17.6% of the contracts of $H_q(I_T)$, the backlink ordering 16.1%, C-PR′ 14.4% and the breadth-first crawl 10.9%, where a random order reaches 10.0%. On $H_q(I_A)$ the backlink ordering had reached 24.7%, AWP′ 21.2%, the breadth-first crawl 17.8% and C-PR′ 13.7%: the C-PR′ curve leads the breadth-first curve over the first visits and falls behind it before 10% of the universe is visited, which is why D_1 is small on this importance. The crawls ordered by a walk depend little on the seed, the breadth-first and backlink crawls much: over the 20 seeds $\mathrm{AUC}_{10\%}$ on I_T runs from 0.1026 to 0.1125 for C-PR′ and from 0.0004 to 0.1263 for the breadth-first crawl.

Discovery order: outside the decision

Table 3 gives the contrasts that the protocol reports without a verdict, in its order. First, each walk against the count it smooths. AWP′ lies above the transfer backlink count on both importances, by +0.0329 [+0.0298, +0.0368] on I_T and +0.0116 [+0.0071, +0.0162] on I_A, and it has the highest $\mathrm{AUC}_{10\%}$ of the orderings compared,

Fig 1. Discovery curves of the deciding run, and the single-layer walks against the counts they smooth out of window. (A) Discovery-order test at T_{obs} in the `both` mode (universe of 1,169,044 contracts): share of the 13,141 hot contracts by user-account inflow, I_T, that a crawl has visited, against the contracts visited. Blue solid line: crawl ordered by the coupled walk C-PR′. Orange dashed line: crawl ordered by the number of contract backlinks, IB'_U. Green dash-dotted line: crawl ordered by the transfer walk AWP′. Thin dark line: breadth-first crawl. Lines are means over the 20 seeds and shaded bands the range over the seeds. Grey dotted line: random curve. Black dotted line near the vertical axis: ideal order. The vertical line marks 10% of the universe, up to which the area of the statistic is taken. (B) The same for the 11,662 hot contracts by user-authorized value, I_A. (C) Spender contracts, out of window: each walk minus the raw count it smooths, as a paired difference in Kendall's τ_b with its unadjusted 95% bootstrap interval (400 resamples). Blue circles: W1, freeze 30 June 2026, labels July–September 2026 ($n = 7{,}915$). Orange squares: W0, freeze 31 March 2026, labels April–June 2026 ($n = 7{,}864$). The two lowest rows restrict W0 to the 4,569 spenders that another contract had paid before the freeze, a subset chosen after the results were known. All differences in (C) were computed after the labels were known.

0.1465 and 0.1582; only the diagnostic δ' (0.1643 and 0.1942) and, among the references that are not crawls, the ideal order and the order by union in-degree of the full graph (0.1593 and 0.1994) lie higher. The walks on the allowance layer alone do not rise above the allowance backlink count on I_T: EndorseRank′ and C-PR′(λ=1) lie below it on both importances by less than 0.002, and EndorseRank-activity′ is not resolved from it on I_T (-0.0014 [$-0.0030, +0.0001$]) and lies above it on I_A ($+0.0161$ [$+0.0136, +0.0185$]). The transfer layer of the coupled walk walked alone, C-PR′(λ=0), lies below the transfer backlink count (-0.0074 [$-0.0109, -0.0031$] and -0.0660 [$-0.0709, -0.0609$]), and C-PR′ lies above the allowance backlink count and below the transfer backlink count on both importances.

Second, the walks among themselves. The allowance layer adds a small, resolved amount to the coupled walk: C-PR′ minus C-PR′(λ=0) is $+0.0024$ [$+0.0016, +0.0033$] on I_T and $+0.0137$ [$+0.0117, +0.0156$] on I_A. C-PR′ lies below AWP′ by -0.0379 [$-0.0412, -0.0347$] and -0.0639 [$-0.0683, -0.0591$]. The two walks differ in the weights and restarts of their transfer layer as well as in the allowance layer, and the contrast does not say which of these matters. Third, the other values of λ give the same reading: at 0.25 and at 0.75 both intervals of D_1 lie above zero and both intervals of D_2 below it. Fourth, on the hot sets defined on the edges that the orderings read, each ordering finds its own target. On the hot set of the union in-degree the backlink ordering is far sooner than C-PR′ (-0.3031 [$-0.3106, -0.2958$]), and on the hot set of C-PR of the full graph C-PR′ is far sooner than the backlink ordering ($+0.2606$ [$+0.2544, +0.2663$]). The first of these is the comparison nearest to the web experiment, where the PageRank ordering reached the pages with many backlinks sooner than the backlink ordering [14]; here it has the opposite sign, in this run and in the three others (S1 Table).

Other settings, the other reveal mode and the other freeze

Table 4 gives the two decision contrasts under the protocol's sensitivities. They change single cells and leave one reading in place: in the `both` mode C-PR′ does not reach the hot contracts sooner than the backlink ordering. Within the `both` mode at T_{obs}, the interval of D_2 lies below zero in every cell but two, the area over 5% of the visits on I_T ($+0.0026$ [$-0.0010, +0.0064$]) and the bootstrap over the seeds on I_T (-0.0051 [$-0.0249, +0.0179$]), where it includes zero. The lead over the breadth-first crawl is firm on I_T and fragile on I_A. On I_A, D_1 is below zero when the area is taken over 20% of

Table 3. Discovery-order test: the contrasts reported outside the decision, in the order of the protocol.

Contrast	On $H_q(I_T)$	On $H_q(I_A)$
1. Each walk′ against the count it smooths		
EndorseRank′ − IB'_A	−0.0006 [−0.0007, −0.0004]	−0.0012 [−0.0014, −0.0010]
EndorseRank-activity′ − IB'_A	−0.0014 [−0.0030, +0.0001]	+0.0161 [+0.0136, +0.0185]
C-PR′ ($\lambda = 1$) − IB'_A	−0.0009 [−0.0010, −0.0007]	−0.0017 [−0.0019, −0.0015]
AWP′ − IB'_T	+0.0329 [+0.0298, +0.0368]	+0.0116 [+0.0071, +0.0162]
C-PR′ ($\lambda = 0$) − IB'_T	−0.0074 [−0.0109, −0.0031]	−0.0660 [−0.0709, −0.0609]
C-PR′ − IB'_A	+0.0459 [+0.0427, +0.0492]	+0.0062 [+0.0023, +0.0106]
C-PR′ − IB'_T	−0.0050 [−0.0086, −0.0008]	−0.0523 [−0.0575, −0.0468]
2. Among the walks		
C-PR′ − C-PR′ ($\lambda = 0$)	+0.0024 [+0.0016, +0.0033]	+0.0137 [+0.0117, +0.0156]
C-PR′ − AWP′	−0.0379 [−0.0412, −0.0347]	−0.0639 [−0.0683, −0.0591]
C-PR′ − C-PR′ ($\lambda = 1$)	+0.0468 [+0.0436, +0.0501]	+0.0078 [+0.0040, +0.0123]
C-PR′ − EndorseRank′	+0.0465 [+0.0433, +0.0498]	+0.0074 [+0.0035, +0.0117]
S-PR′ − C-PR′ ($\lambda = 0$)	+0.0008 [+0.0006, +0.0011]	+0.0054 [+0.0044, +0.0066]
3. Other values of λ		
C-PR′ ($\lambda = 0.25$) − BFS	+0.0515 [+0.0480, +0.0555]	+0.0049 [+0.0002, +0.0100]
C-PR′ ($\lambda = 0.25$) − IB'_U	−0.0055 [−0.0091, −0.0016]	−0.0584 [−0.0633, −0.0531]
C-PR′ ($\lambda = 0.25$) − C-PR′ ($\lambda = 0.5$)	−0.0005 [−0.0008, −0.0001]	−0.0012 [−0.0017, −0.0008]
C-PR′ ($\lambda = 0.75$) − BFS	+0.0526 [+0.0491, +0.0563]	+0.0070 [+0.0024, +0.0121]
C-PR′ ($\lambda = 0.75$) − IB'_U	−0.0045 [−0.0080, −0.0005]	−0.0562 [−0.0610, −0.0508]
C-PR′ ($\lambda = 0.75$) − C-PR′ ($\lambda = 0.5$)	+0.0006 [+0.0002, +0.0010]	+0.0009 [+0.0005, +0.0014]
4. Hot sets in the manner of Cho et al. (an intended-construct check)		
C-PR′ − BFS on $H_q(\mathrm{IB}_U)$ (13,683 contracts)	+0.0176 [+0.0119, +0.0238]	
C-PR′ − IB'_U on $H_q(\mathrm{IB}_U)$ (13,683 contracts)	−0.3031 [−0.3106, −0.2958]	
C-PR′ − BFS on $H_q(\mathrm{IR})$ (11,691 contracts)	+0.3084 [+0.3024, +0.3143]	
C-PR′ − IB'_U on $H_q(\mathrm{IR})$ (11,691 contracts)	+0.2606 [+0.2544, +0.2663]	

Deciding run (T_{obs}, `both` mode); the rows follow the order of the protocol's list. Each cell is a difference of $\mathrm{AUC}_{10\%}$ between two orderings with the 95% percentile interval of the paired bootstrap over the hot contracts, 400 resamples (seed 42); an interval above 0 means that the first ordering reached the hot contracts sooner. C-PR′ without a value of λ is C-PR′ at $\lambda = 0.5$; C-PR′ ($\lambda = 1$) and C-PR′ ($\lambda = 0$) are its allowance and transfer layers walked alone. Block 1 sets each walk beside the backlink count of the layer it walks. In block 2, C-PR′ against C-PR′ ($\lambda = 0$) shows what the allowance layer adds to the coupled walk; C-PR′ and AWP′ differ in the weights and restarts of their transfer layer as well. In block 4 the hot sets are defined on the edges that the orderings read, the union in-degree IB_U and C-PR of the full graph (IR), so agreement there is an intended-construct check; C-PR′ minus IB'_U on $H_q(\mathrm{IB}_U)$ is the contrast that the protocol calls the replication contrast. None of these rows carries a verdict. Source: `discovery/discovery_tobs_both.json`, key `fixed_order` (items 1 to 4).

the visits (−0.0518 [−0.0575, −0.0460]) and when the ordering is recomputed four times as often (−0.0134 [−0.0181, −0.0082]), and its interval includes zero without the hubs (−0.0019 [−0.0065, +0.0026]) and under the bootstrap over the seeds (+0.0061 [−0.0315, +0.0398]).

The reveal mode changes the result. In the `out` mode, where a visit reveals out-edges only and the universe is the SCC with OUT, all four contrasts lie above zero at T_{obs}: D_1 is +0.2650 [+0.2540, +0.2743] on I_T and +0.1143 [+0.1053, +0.1237] on I_A, and D_2 is +0.0803 [+0.0690, +0.0920] and +0.0436 [+0.0347, +0.0522]. This mode is a sensitivity analysis under the protocol's rule and carries no verdict. In it the walks on the transfer layer also lie above the transfer backlink count, and coupling lowers the area on I_T and raises it on I_A (S1 Table). The diagnostics point to ties as one difference between the modes. The count orderings decide nearly every pick by a tie in both modes (more than 0.999 of the picks). C-PR′ decides 0.535 of its picks by a tie in the `both` mode and 0.023 in the `out` mode, and AWP′ 0.252 and 0.005. At the other freeze the two modes read as at T_{obs}: at t_1 in the `both` mode D_1 is above zero and D_2 below zero on both importances, and in the `out` mode all four contrasts are above zero (Table 4). At t_1 in the `both` mode AWP′ again lies above the transfer backlink count on both importances, EndorseRank-activity′ lies above the allowance backlink count on both, and the backlink ordering IB'_U has the highest area on I_A (S1 Table).

The protocol reports one further reading as curves only, without an interval: the

Table 4. Discovery-order test: the two decision contrasts under other settings, in the other reveal mode and at the other freeze.

Setting	Contrast	On $H_q(I_T)$	On $H_q(I_A)$
The decision (T_{obs}, `both` mode, $x = 10\%$, $q = 1\%$, $\rho = 1/50$)			
As decided	D_1	+0.0520 [+0.0485, +0.0558]	+0.0061 [+0.0014, +0.0111]
	D_2	−0.0051 [−0.0087, −0.0010]	−0.0571 [−0.0621, −0.0519]
5. Sensitivities of the deciding run, one setting varied at a time			
Area over $x = 5\%$ of the visits	D_1	+0.0673 [+0.0635, +0.0714]	+0.0472 [+0.0429, +0.0521]
	D_2	+0.0026 [−0.0010, +0.0064]	−0.0132 [−0.0178, −0.0084]
Area over $x = 20\%$ of the visits	D_1	+0.0199 [+0.0169, +0.0230]	−0.0518 [−0.0575, −0.0460]
	D_2	−0.0063 [−0.0091, −0.0034]	−0.0966 [−0.1023, −0.0905]
Hot share $q = 5\%$[a]	D_1	+0.0132 [+0.0121, +0.0144]	+0.0061 [+0.0014, +0.0111]
	D_2	−0.0067 [−0.0078, −0.0057]	−0.0571 [−0.0621, −0.0519]
Hot sets without the 1,170 hubs[b]	D_1	+0.0306 [+0.0274, +0.0341]	−0.0019 [−0.0065, +0.0026]
	D_2	−0.0126 [−0.0159, −0.0088]	−0.0607 [−0.0654, −0.0553]
Importance by counts (N_T, N_A)[c]	D_1	+0.0188 [+0.0163, +0.0211]	+0.0191 [+0.0129, +0.0262]
	D_2	−0.0282 [−0.0308, −0.0254]	−0.0473 [−0.0537, −0.0408]
Bootstrap over the seeds	D_1	+0.0520 [+0.0320, +0.0717]	+0.0061 [−0.0315, +0.0398]
	D_2	−0.0051 [−0.0249, +0.0179]	−0.0571 [−0.0873, −0.0221]
Recomputed four times as often ($\rho = 1/200$, 5 seeds)	D_1	+0.0381 [+0.0347, +0.0418]	−0.0134 [−0.0181, −0.0082]
	D_2	−0.0129 [−0.0167, −0.0087]	−0.0713 [−0.0765, −0.0658]
The other reveal mode (`out`) at T_{obs}[d]	D_1	+0.2650 [+0.2540, +0.2743]	+0.1143 [+0.1053, +0.1237]
	D_2	+0.0803 [+0.0690, +0.0920]	+0.0436 [+0.0347, +0.0522]
6. The other freeze, t_1			
`both` mode[d]	D_1	+0.0574 [+0.0530, +0.0611]	+0.0230 [+0.0164, +0.0291]
	D_2	−0.0158 [−0.0196, −0.0121]	−0.1010 [−0.1083, −0.0933]
`out` mode[d]	D_1	+0.3414 [+0.3282, +0.3547]	+0.1705 [+0.1564, +0.1822]
	D_2	+0.1222 [+0.1079, +0.1361]	+0.0739 [+0.0607, +0.0848]

Differences of AUC with 95% percentile intervals: D_1 is C-PR′ minus BFS and D_2 is C-PR′ minus IB'_U. Unless a row says otherwise the interval is that of the paired bootstrap over the hot contracts, 400 resamples (seed 42). The decision is read in the first block only; no other row carries a verdict. [a]Only 11,662 contracts of the universe have a positive I_A, fewer than 1% of it, so the hot set of I_A is the same at both shares. [b]The contracts with the highest union in-degree of the full graph, taken out of the hot sets and kept in the graph. [c]Hot sets by the number of user accounts that transferred, or had authorized, at least one US dollar. [d]In the `out` mode a visit reveals out-edges only and the universe is the giant strongly connected component with the contracts reached from it. Universes and hot sets (I_T and I_A) of the three runs outside the decision: T_{obs}, `out`: 508,637 contracts, 5,087 and 5,088 hot contracts; t_1, `both`: 1,015,867 contracts, 12,697 and 8,136 hot contracts; t_1, `out`: 332,396 contracts, 3,324 and 3,324 hot contracts. The two freezes share nine months of their score windows and are not independent. Sources: `discovery/discovery_tobs_both.json` (`decision`, `fixed_order.5_sensitivities`), `discovery_tobs_out.json`, `discovery_t1_both.json` and `discovery_t1_out.json` (`decision`).

share of the summed importance of the universe that the visited contracts hold. By value the walks lead early. After 1% of the visits of the deciding run the contracts visited by the crawl ordered by C-PR′ held 71.4% of the summed I_T and 84.1% of the summed I_A; for the backlink ordering the shares were 42.1% and 47.7%, for AWP′ 80.0% and 46.4%, and for the breadth-first crawl 7.6% and 6.0%. After 10% of the visits the backlink ordering had caught up with C-PR′ (74.3% and 85.9% against 73.5% and 87.0%), and AWP′ held 82.0% of I_T but 49.2% of I_A, the lowest share of the four. User-account demand is thus concentrated in few contracts, which the walks reach within the first visits, while the hot sets, which give each of their thousands of contracts the same weight, are reached slowly by every ordering: in the first 10% of the visits between 0.835 and 0.911 of the visits of the counts and walks went to contracts with no user-account demand at all (S1 Table).

Each score agrees most with the checks of its own edges

In the same window each single-layer score follows the edges it walks on (Table 5). On the allowance family EndorseRank reaches $\tau = 0.839$ $[0.828, 0.849]$ against 0.227 $[0.206, 0.245]$ for AWP; on the transfer family AWP reaches 0.755 $[0.748, 0.762]$ against

0.297 [0.280, 0.317] for EndorseRank. The paired differences are $\Delta\tau = +0.613$ [0.591, 0.632] (EndorseRank minus AWP, allowance family) and +0.458 [0.438, 0.476] (AWP minus EndorseRank, transfer family). This is the pattern an intended-construct check expects, and part of it is built in: EndorseRank agrees most with the in-approve degree (0.847 [0.835, 0.857]), the count it smooths, and both are zero for the cohort contracts outside the allowance layer. With dollar amounts, AWP's agreement with the value of inbound transfers (0.746 [0.738, 0.754]) comes close to its agreement with the number of distinct senders (0.764 [0.757, 0.772]), as its value weight intends. On the Sybil-stability family, which averages transfer-based measures of how long and how evenly a contract has received transfers, AWP reaches 0.643 [0.635, 0.651] and EndorseRank 0.220 [0.206, 0.237]. With activity restarts EndorseRank agrees less with its own family (0.799 [0.782, 0.813]) and about as much with the transfer family (0.299 [0.281, 0.318]).

Table 5. Same-window agreement of each score with the checks of each edge type (intended-construct check).

Check (Kendall τ_b [95% CI])	EndorseRank	EndorseRank, activity restarts	AWP
Transfer family (mean)	0.297 [0.280, 0.317]	0.299 [0.281, 0.318]	0.755 [0.748, 0.762]
in-degree	0.305 [0.287, 0.326]	0.306 [0.288, 0.327]	0.764 [0.757, 0.772]
in-value (USD)	0.289 [0.269, 0.309]	0.291 [0.271, 0.310]	0.746 [0.738, 0.754]
Allowance family (mean)	0.839 [0.828, 0.849]	0.799 [0.782, 0.813]	0.227 [0.206, 0.245]
in-approve degree	0.847 [0.835, 0.857]	0.814 [0.798, 0.828]	0.229 [0.207, 0.248]
in-approve value (USD, capped)	0.831 [0.820, 0.841]	0.783 [0.767, 0.797]	0.225 [0.204, 0.243]
Sybil-stability family (mean)	0.220 [0.206, 0.237]	0.225 [0.210, 0.242]	0.643 [0.635, 0.651]
Paired differences $\Delta\tau$ [95% CI] and agreement of the two rankings			
EndorseRank − AWP, allowance family	+0.613 [0.591, 0.632]		
AWP − EndorseRank, transfer family	+0.458 [0.438, 0.476]		
τ_b between EndorseRank and AWP	0.275 [0.258, 0.294]		
pairs of contracts ordered by both scores	6,704,900 of 20,752,903 (32.3%); 26.7% of them discordant		
Contracts outside a graph kept as isolated nodes (the sensitivity analysis fixed in the plan)			
EndorseRank, allowance family	0.956 [0.952, 0.960]		
AWP, transfer family	0.755 [0.748, 0.762]		
τ_b between EndorseRank and AWP	0.227 [0.207, 0.245]		

Matched cohort, $n = 6{,}443$ contracts, scores at T_{obs} (30 Jun 2026) from the score window 1 Jul 2025 to 30 Jun 2026; the checks are computed over the same window from the events between contracts. Each check is computed from the events of one layer: the transfer family from inbound transfers, the allowance family from the capped latest allowances received. Agreement of a score with the checks of its own edge type is therefore partly built in. The Sybil-stability family averages the inbound-counterparty ratio, transfer tenure and active months. Family means are plain averages of the coefficients of their checks. A contract outside a graph scores zero: 5,200 cohort contracts score zero under EndorseRank and 2,384 under AWP (5,193 and 2,384 lie outside the allowance and the transfer graph, 2,173 outside both), so the scores tie on these contracts. Paired contract bootstrap, 400 resamples (seed 42); intervals are unadjusted. None of these comparisons decides a rule. Source: `same-window/eval_summary.json`, keys `families`, `per_proxy`, `contrasts`, `inter_method`, `isolated_rule`, `ties`, `graph`.

The two walks order the contracts differently. Their rankings agree at $\tau_b = 0.275$ [0.258, 0.294]. Only 32.3% of the 20,752,903 pairs of cohort contracts are ordered by both scores, because most cohort contracts score zero under EndorseRank, and the two scores disagree on the direction for 26.7% of the pairs that both order. Keeping the contracts outside a graph as isolated nodes, which the plan fixed as a sensitivity analysis, raises EndorseRank's agreement with the allowance family to 0.956 [0.952, 0.960], leaves AWP's coefficients as they are and lowers the agreement of the two rankings to 0.227 [0.207, 0.245].

Out of window: each walk against the count it smooths

Table 6 sets every score frozen at each date against the two spender labels. Two facts frame every number in it: the labels count mostly user accounts, which no score reads, and about 27% of each spender cohort lies outside the union graph and scores zero. The coefficients are accordingly small. The edge type still decides which label a score ranks

best. In W0 the in-approve degree ranks new approval pairs at $\tau = 0.204$ $[0.181, 0.228]$ and EndorseRank at 0.168 $[0.145, 0.193]$, while the transfer in-degree (0.070 $[0.050, 0.091]$) and AWP (0.098 $[0.079, 0.119]$) stay lower. On new transfer senders the order reverses: AWP reaches 0.330 $[0.313, 0.347]$ and the transfer in-degree 0.318 $[0.298, 0.336]$, against 0.084 $[0.062, 0.105]$ for EndorseRank. EndorseRank minus AWP is $\Delta\tau = +0.070$ $[0.042, 0.096]$ on new approval pairs and -0.246 $[-0.271, -0.220]$ on new senders, and W1 repeats the pattern with $+0.083$ $[0.059, 0.110]$ and -0.250 $[-0.273, -0.225]$.

Table 6. Spender holdouts: scores frozen at each date against labels counted over the next three months.

	W0: freeze 31 Mar 2026 (n = 7,864)		W1: freeze 30 Jun 2026 (n = 7,915)	
Score at the freeze	New approval pairs	New senders	New approval pairs	New senders
In-approve degree (count)	0.204 [0.181, 0.228]	0.091 [0.066, 0.113]	0.181 [0.158, 0.208]	0.072 [0.046, 0.096]
Transfer in-degree (count)	0.070 [0.050, 0.091]	0.318 [0.298, 0.336]	0.031 [0.008, 0.052]	0.302 [0.278, 0.323]
EndorseRank	0.168 [0.145, 0.193]	0.084 [0.062, 0.105]	0.147 [0.125, 0.170]	0.064 [0.041, 0.085]
EndorseRank, activity restarts	0.163 [0.142, 0.188]	0.089 [0.069, 0.111]	0.148 [0.126, 0.171]	0.072 [0.049, 0.093]
AWP	0.098 [0.079, 0.119]	0.330 [0.313, 0.347]	0.064 [0.042, 0.084]	0.314 [0.295, 0.332]
Each walk against the count it smooths, on the label of its own edge type: $\Delta\tau$ [95% CI]				
EndorseRank − in-approve degree	−0.036 [−0.046, −0.027]		−0.034 [−0.043, −0.025]	
EndorseRank, activity restarts − in-approve degree	−0.041 [−0.053, −0.029]		−0.033 [−0.044, −0.022]	
AWP − transfer in-degree		+0.012 [0.003, 0.023]		+0.011 [0.001, 0.022]
The same in W0 among the 5,700 spenders inside the union graph[a]				
EndorseRank − in-approve degree	−0.041 [−0.051, −0.031]		–	
AWP − transfer in-degree		+0.021 [0.005, 0.038]		–
The same in W0 among the 4,569 spenders that another contract had paid before the freeze[a]				
EndorseRank − in-approve degree	−0.045 [−0.058, −0.033]		–	
AWP − transfer in-degree		−0.035 [−0.049, −0.022]		–
Allowance walk against transfer walk: $\Delta\tau$ [95% CI]				
EndorseRank − AWP	+0.070 [0.042, 0.096]	−0.246 [−0.271, −0.220]	+0.083 [0.059, 0.110]	−0.250 [−0.273, −0.225]

Kendall τ_b with 95% percentile interval from 400 paired contract resamples (seed 42); intervals are unadjusted. Scores and counts are computed on the contract graph of the freeze. Spenders are the contracts that hold a positive latest allowance of one of the five tokens from any account, set in the score window, at the freeze. The labels count every account that newly approves or pays a spender, user accounts included: in W0 90.4% of the 748,258 new approval pairs and 86.3% of the 651,474 new senders are user accounts, in W1 87.9% of 702,275 and 82.3% of 618,461. In W0, 1,725 spenders gained at least one new approval pair and 1,416 at least one new transfer sender; in W1, 1,491 and 1,369 did. 2,164 W0 spenders (27.5%) and 2,156 W1 spenders (27.2%) end no pair between contracts and score zero under every score and count. Every paired difference in this table was computed after the labels of its window were known, and none of them decides a rule. [a]Subsets chosen after the results were known; – marks contrasts that were not computed for W1. Sources: `holdout-w0/eval_summary.json` (`taus`, `contrasts_after`, `receiving`, `in_union_graph`, `label_split`, `cohort_coverage`), `window-w1/eval_summary.json` (`spender_taus`, `endorserank_results`, `label_split`, `cohort_coverage`), `supplementary/supplementary.json`.

On the allowance side the walk loses to its count (Fig 1C). EndorseRank, restarting uniformly as the plan fixed, trails the in-approve degree on later new approval pairs by $\Delta\tau = -0.036$ $[-0.046, -0.027]$ in W0 and -0.034 $[-0.043, -0.025]$ in W1. With activity restarts it trails the count likewise, by -0.041 $[-0.053, -0.029]$ and -0.033 $[-0.044, -0.022]$; the restart rule makes little difference to the walk on this label (-0.004 $[-0.008, -0.001]$ in W0 and $+0.001$ $[-0.002, 0.005]$ in W1 for activity minus uniform restarts).

On the transfer side the answer goes both ways. On the whole spender cohorts AWP ranks later new transfer senders slightly above the transfer in-degree, by $+0.012$ $[0.003, 0.023]$ in W0 and $+0.011$ $[0.001, 0.022]$ in W1, and among the 5,700 W0 spenders inside the union graph by $+0.021$ $[0.005, 0.038]$. Among the 4,569 W0 spenders that another contract had paid before the freeze, a subset chosen after the results were known, AWP lies below the count, by -0.035 $[-0.049, -0.022]$, and EndorseRank stays below the in-approve degree there (-0.045 $[-0.058, -0.033]$). The two results are compatible. The transfer in-degree is zero for every spender that no other contract has paid, while AWP is zero only for the 2,421 W0 spenders without any transfer edge;

AWP thus gives a positive score to spenders that sent transfers and received none, which the count ties with the spenders outside the layer. Where every spender has a positive in-degree, the count ranks the later senders better. Neither the plan nor the registration document named any of these contrasts. In the registered analysis the evaluation code committed with the registration, before this study extracted the W1 logs, already contained the two W1 contrasts of EndorseRank and AWP with their degrees; the W0 contrasts, the activity-variant contrasts and every contrast of the main analysis were chosen and computed after the labels were known.

Two further labels, which the plan did not name, were computed on the W0 spenders: revocations, the owners of a positive allowance at the freeze that set it to zero in the label window, and drained owners, a heuristic count of owners whose new approval was followed within hours by an outgoing transfer of the same token (S1 Text). Every score ranks both positively. Revocations follow the in-approve degree most closely (0.190 [0.161, 0.220]) and the transfer in-degree least (0.061 [0.040, 0.083]); for drained owners the coefficients run from 0.092 [0.072, 0.113] for the transfer in-degree to 0.213 [0.188, 0.238] for the in-approve degree. A contract that owners trusted more should lose approvals less often, not more, so these counts fit the volume of owner activity rather than endorsement, and neither is offered as evidence.

What EndorseRank computes on depth-one spenders

In the allowance layer most contracts are either owners or spenders, and a spender whose owner contracts receive no allowance themselves sits at depth one. For such a spender EndorseRank can be written out.

Observation. Let PageRank follow Eq (2) with damping d and uniform restarts, $\pi_u = 1/|V|$, and let r_D be the total score of the dangling nodes. Every node without in-edges then has the common score

$$c = \frac{1 - d + d\, r_D}{|V|}, \tag{10}$$

and a node v none of whose in-neighbors has an in-edge of its own scores

$$r_v = c\,\big(1 + d\,\delta(v)\big), \qquad \delta(v) = \sum_u \frac{w(u,v)}{o(u)}. \tag{11}$$

Here $\delta(v)$ is the owner-normalized in-strength of v: each owner u contributes the share of its own approval weight that goes to v, so an owner that approves many spenders adds little to each of them.

Derivation. At the fixed point of Eq (2) every node v receives $(1-d)/|V|$ from teleportation and $d\, r_D/|V|$ from the dangling rows, together c, plus $d \sum_u r_u\, w(u,v)/o(u)$ from its in-neighbors. A node without in-edges receives only c. If every in-neighbor u of v is such a node, substituting $r_u = c$ gives Eq (11).

The expression is a one-step expansion of the fixed point rather than a closed form in the strict sense, because c depends on r_D, which the whole score vector determines; but c is common to all spenders it covers, so EndorseRank orders those spenders exactly as $\delta(v)$ does. It describes the computed scores (Table 1). In the graph at T_{obs} every node without in-edges has the same score, $c = 8.45 \times 10^{-6}$, up to floating-point error. Among the 926 cohort contracts with an allowance in-edge, the 651 of depth one all lie within 1% of Eq (11); among the remaining 275, whose owners include at least one contract that is itself approved, 33.8% do, which makes 80.3% over all spenders. On the graph at t_1, 810 of the 1,362 W0 spenders with an allowance in-edge have depth one, again all within 1%, and 68.4% of these W0 spenders lie within 1%.

This has two consequences. First, for a depth-one spender the walk does nothing but reweight its approving contracts: an approval weighs more when it is recent, and less when its owner spreads its approvals over many spenders. Apart from these weights and this division, EndorseRank and the count carry the same information about such a spender, and there is nothing further for propagation to supply. The division matters most where one owner approves very many spenders, as the largest owner contract at T_{obs} does. Second, longer paths exist: 275 of the 926 cohort spenders at T_{obs} and 552 of the 1,362 W0 spenders are approved by an owner contract that others approve in turn, and most of these scores differ from the closed form by more than 1%. Propagation over several steps therefore occurs in the allowance layer; it did not lift EndorseRank above the raw count out of window, with uniform or with activity restarts.

Computational cost

With one solver and identical settings, EndorseRank's solver graph (114,997 edges on 109,381 nodes) took 0.101 s on average and AWP's (2,580,381 edges on 1,298,649 nodes) 2.821 s, over five timed runs with standard deviations of 0.009 and 0.189 s. AWP thus took 27.90 times as long at an edge ratio of 22.44; it also needed more iterations (98 against 56), and its peak memory was 228.2 MB against 13.7 MB. Matrix assembly and each power iteration are linear in the number of edges [59], so the difference in time follows from the size of the transfer layer and from the iterations the walk needs on it, which the benchmark does not separate, and not from any property of the algorithm. A walk that also uses the allowance edges pays little more: the coupled walk over both layers (2,591,253 pairs) took 3.080 s.

Ordering a crawl by a walk multiplies that cost. In the deciding run every ordering was recomputed about 540 times per crawl. A count needs no solve. A crawl ordered by C-PR′ ran 540 solves with 22,594 power iterations on average, and one ordered by AWP′ 542 solves with 29,042, on graphs that grow to the size of the universe. The whole deciding run, with its 15 orderings, 20 seeds and sensitivities, took 29,948 seconds on 16 parallel workers.

The registered analysis

Table 7 sets the registered analysis beside the main one. The registered analysis kept amounts in raw base units over every ERC-20 token, scored every event since 1 October 2023 and built graphs that hold every account of the cohort's events, with the plan's cohort of 27,844 contracts. No rule of the plan or the registration decides any quantity of this paper, so the comparison is descriptive. In both analyses EndorseRank fell below the in-approve degree on new approval pairs in both windows, AWP rose above the transfer in-degree on new senders on the whole spender cohorts in both windows, and EndorseRank met the closed form for every depth-one spender. They differ on the variant with activity restarts, which rose above the in-approve degree in the registered analysis (+0.043 [0.036, 0.049] in W0 and +0.018 [0.012, 0.023] in W1) and lies below it in the main analysis. The same-window coefficients are higher in the main analysis; there the scores and the checks are zero together for the many cohort contracts outside a graph, which may account for part of the rise. The discovery-order test has no registered counterpart.

Table 7. The registered analysis beside the main analysis.

	Registered analysis	Main analysis (contract graph)
Definitions		
Amounts; tokens	raw base units; every ERC-20	US dollars; five tokens
Events behind the scores at T_{obs}	1 Oct 2023 – 30 Jun 2026	1 Jul 2025 – 30 Jun 2026
Events behind the scores at t_1	1 Oct 2023 – 31 Mar 2026	1 Apr 2025 – 31 Mar 2026
Nodes of the graphs	every account of the cohort's events	contracts other than smart accounts
Events read for the edges	with a cohort contract at one end	between contracts, whole chain
Each latest allowance counted	in full	up to the owner's balance
Same window		
Matched cohort (contracts)	27,844	6,443
Allowance / transfer edges at T_{obs}	22,321,470 / 59,737,617	114,997 / 2,580,381
EndorseRank, allowance family	0.481 [0.476, 0.487]	0.839 [0.828, 0.849]
AWP, transfer family	0.447 [0.440, 0.453]	0.755 [0.748, 0.762]
τ_b between EndorseRank and AWP	0.217 [0.209, 0.225]	0.275 [0.258, 0.294]
Depth-one / all spenders within 1% of Eq (11)	100.0% / 86.4%	100.0% / 80.3%
Solver time / edges, AWP over EndorseRank	2.36 / 2.68	27.90 / 22.44
Spender holdouts, each walk minus the count it smooths (computed after the labels)		
Spender cohorts, W0 / W1	33,101 / 34,587	7,864 / 7,915
W0: EndorseRank − in-approve degree, new pairs	−0.011 [−0.016, −0.006]	−0.036 [−0.046, −0.027]
W0: EndorseRank (activity) − in-approve degree	+0.043 [0.036, 0.049]	−0.041 [−0.053, −0.029]
W0: AWP − transfer in-degree, new senders	+0.038 [0.035, 0.042]	+0.012 [0.003, 0.023]
W1: EndorseRank − in-approve degree, new pairs	−0.029 [−0.032, −0.025]	−0.034 [−0.043, −0.025]
W1: EndorseRank (activity) − in-approve degree	+0.018 [0.012, 0.023]	−0.033 [−0.044, −0.022]
W1: AWP − transfer in-degree, new senders	+0.020 [0.016, 0.023]	+0.011 [0.001, 0.022]
Discovery-order test		
Verdict on I_T and on I_A	no registered counterpart	V2 and V2 (Table 2)

The registered analysis follows the analysis plan and the W1 registration as committed: amounts in raw base units of every ERC-20 token, scores from every event since 1 Oct 2023 and graphs that hold every account of the cohort's events. The main analysis is the contract-graph analysis; it was defined after the registered results were known, so no value of its column is a registered result. No rule of the plan or the registration decides any row of this table: the count contrasts were computed after the labels of their windows were known in both analyses, and the discovery-order test belongs to the main analysis only. $\Delta\tau$ and τ_b with 95% interval from 400 paired contract resamples (seed 42). The ratios of solver time and of edges divide the values of AWP's graph by those of EndorseRank's graph in each analysis. Sources: `same-window`, `holdout-w0`, `registered-w1` and `window-w1` summaries, `supplementary.json` and `benchmark.json` of each analysis; `analysis/closed-form/closed_form_check.json`; the configuration files for the windows.

Discussion

What the test shows and what it does not

The test asked a question with a yes or no answer, and the answer was no. In the deciding run a crawl ordered by the coupled walk reached the contracts with the most user-account demand sooner than a breadth-first crawl, but a crawl ordered by a plain count, the number of contracts already visited that pay or authorize the candidate, reached them sooner than the walk on both importances. The sensitivities left that reading in place, and at the other freeze the same mode gave the same signs. On the contract graph of this rollup, with in-edges visible, coupling allowance and transfer edges in one walk did not order a crawl better than counting backlinks.

The test does not show that walks are useless for discovery. One walk beat the count it smooths: the crawl ordered by AWP, a transfer walk with value and time weights and restarts at contracts that sent recently, had the highest area in the deciding run and lay above the transfer backlink count on both importances in all four runs. The coupled walk lay below AWP, and its own transfer layer walked alone lay below the transfer backlink count, so walking the transfer layer was not enough by itself. The two transfer walks differ in their weights and in their restarts, and the test does not separate the two. The result also depends on what a visit reveals. Where a visit shows out-edges only, as on the web, the coupled walk led both comparators at both freezes, under a mode that

the protocol's rule made a sensitivity analysis. In that mode the frontier holds only contracts that visited contracts point to, the counts tie on nearly every pick, and the coupled walk and the transfer walks rarely tie; we did not test whether this is the reason. Even there, the comparison nearest to the web experiment did not come out as on the web: on the hot set of contracts with the most backlinks, the backlink ordering was sooner than the coupled walk in all four runs. The contract graph differs from a university's web in ways that may matter, among them a hub that ends 441,612 transfer pairs, seeds drawn by hash and not a home page, and weighted layers; the test does not say which of these accounts for the difference.

Two readings of the same crawls should be held together. Counted over the hot sets, every ordering was far from the ideal order and not far from a random one: after a tenth of the visits no crawl in the comparison had reached more than a quarter of either hot set. Weighted by dollars, the coupled walk had visited contracts that hold most of the inflow and most of the authorized value of the universe within the first hundredth of the visits, sooner than the backlink ordering, which caught up within the first tenth; AWP reached most of the inflow as early but less than half of the authorized value. The demand of user accounts is concentrated in a few contracts that the walks find within the first visits, and the long remainder of the hot sets is what the statistic of the test measures. The value-weighted reading is reported by the protocol as curves only and has no interval.

The allowance layer between contracts is thin, and the test says little in its favor. Under the balance cap most allowances between contracts count for nothing, because the authorizing contract holds no balance of the token at the freeze. Only 16,776 capped allowance pairs lie in the universe of the deciding run against 2,383,503 transfer pairs, the walks on the allowance layer alone were not above the allowance backlink count on inflow, and adding the layer to the transfer layer raised the area by 0.0024 and 0.0137. By value the layer did more: after a tenth of the visits the orderings that read both layers had visited contracts holding 85.9% to 87.0% of the user-authorized value, the orderings that read allowances only 68.4% to 86.5%, and the orderings that read transfers only about half of it (S1 Table). Contracts authorize one another much less than users authorize contracts, and a graph of contracts sees only the first kind.

The walks against their counts out of window

Out of window the two edge types measure different things: scores built on allowances rank which spenders later gain new approvals, and scores built on transfers which later gain new senders. Propagating along the allowance edge added nothing. EndorseRank fell below the count of approving contracts in both windows, with either restart rule, and the closed form says why this is unsurprising for most spenders: there the walk is the count, with each approval scaled by its recency and divided by its owner's total approval weight. On the transfer side AWP was slightly above its count on the whole cohorts and, in W0, below it among the spenders that a contract had already paid, so its small lead comes from the spenders that the count leaves at zero. That a count is hard to beat fits preferential attachment and link-prediction practice [22, 24]. All of these coefficients are small, the labels count mostly user accounts that the graph does not hold, and about 27% of each cohort scores zero, so the comparisons say more about the order of the scores than about how well any of them predicts.

Practical implications

The results support modest advice for those who must choose which contracts to look at first. A wallet, an explorer or an auditor that reads the chain's logs sees both who paid a contract and whom it paid. In that setting the number of distinct contracts that pay

or authorize a contract ordered a crawl at least as well as the coupled walk, at no solver cost, and a transfer walk with AWP's weights and restarts did somewhat better at the price of a solve at every recompute. An integrator that follows only whom a contract itself pays and authorizes is closer to the other mode, where the coupled walk and the transfer walks led; that result carries no verdict here and should be tested before it is relied on. For ranking which spender contracts will gain new approvers, the count of approving contracts ranked above EndorseRank in both windows. For new senders AWP and the transfer in-degree were close, with the count ahead, in the first window, among contracts that were already paid by others.

None of these orderings finds the contracts that users rely on efficiently. The best crawl was far from the ideal order, and most of the first tenth of the visits of every ordering went to contracts with no user-account demand. The contracts that hold most of the demand by value are few, and the walks on the transfer layer reach most of the inflow within the first hundredth of the visits; the difficulty is the long tail. A count of contract backlinks tells how widely other contracts use a contract, not what the contract will do with an allowance: a phishing contract can be paid and authorized by contracts that its operator deploys, at the cost of a deployment and a balance each, and a new contract scores low whatever its code. A low count is a reason to look closer, and a high count is no reason to sign. None of these scores is a security audit, and whether a score of this kind could later serve as one input to credit decisions about contracts was not tested.

Limitations

The evidence comes from one chain, two freezes that share nine months of their score windows, and five tokens, so other rollups and Layer-1 networks would need new data [3], contracts that are used only through other tokens are outside the graph, and native ether enters only as WETH. The contract-graph analysis and the discovery-order test were defined after the registered results were known, so the tokens, price source, windows, account-type rules, balance cap, constants, importance measures and the test's statistic were all chosen by people who had seen those results; the test decides no rule of the plan or the registration, and its verdict is not a registered result. The result depends on the reveal mode, and the mode that decides was named by a rule on the structure of one freeze.

The allowance layer between contracts is thin. The pilot found a capped allowance in-edge on 0.54% of the contracts of the SCC and OUT at t_1, below the protocol's threshold, and the test went on. At T_{obs} one owner contract holds 98,073 of the 115,006 capped allowance pairs, so the allowance-side results at T_{obs}, namely the same-window checks, W1 and the closed form at T_{obs}, lean on the approvals of a single contract, which the study does not identify; at t_1 the largest owner holds 256 pairs.

Account types are read from code at one block after the end of the data, and the smart-account rules find Safes and ERC-4337 accounts. Kinds they miss stay contracts: after the listed code families, 115,084 typed contracts still carry a `validateUserOp` selector and no paymaster selector without answering a canonical EntryPoint, and contracts that services create for each user, order or payment are not picked out. A contract that destroyed itself before the read counts as a user account. Addresses that sent no transaction and whose only counterparties are transaction senders are not typed and cannot be nodes; in an audit sample of 1,000 of them the reads found contract code other than a smart account's for 54, so some contracts that only user accounts use lie outside the graph. Balances are reconstructed from `Transfer` logs and matched `balanceOf` for 99.5% of a sample, so tokens that change balances without such logs would be misread.

Coverage is partial: about a third of the matched cohort and 27% of each spender

cohort lie outside the union graph and score zero, which ties many members and lowers every coefficient. The labels count events around the cohort and mostly user accounts, which no score reads, and they are sparse counts of arrivals, which favor any score that tracks the current number of counterparties. The importance measures are sums of capped dollar amounts, the hot set of I_A is every contract with a positive value, and the intervals of the test resample hot contracts with the 20 seeds fixed; under a bootstrap over the seeds two of the four decision intervals include zero. The schedule recomputes an ordering only after each growth of the visits by one fiftieth.

Every interval of the holdouts treats contracts as independent and the graphs as fixed, although spenders share owners, so the intervals are probably too narrow; the analyses use only 400 resamples, and no interval is adjusted for multiplicity. AWP's parameters are our choices, as its paper gives none. Approvals older than the score window are not read even when they are still in force, an `Approval` event can record the remainder after a spend as well as a grant, and allowances inside managers such as Permit2 are unseen. No outcome outside the chain was available as an importance label, and no causal claim is made.

Conclusions

Does PageRank find the smart contracts that users rely on sooner than counting? On the contract graph of Arbitrum One the answer depends on the walk and on what a crawl can see, and for the walk on which the test's decision rested it is no. In the run that decides the test, a crawl ordered by the walk that couples allowance and transfer edges reached the contracts with the most user-account demand sooner than a breadth-first crawl but not sooner than a crawl ordered by the number of contract backlinks, which was sooner. In the same run the transfer walk AWP did reach them sooner than the count it smooths, and the walks on allowance edges alone did not on inflow; where a visit reveals out-edges only the coupled walk led both comparators, without a verdict. Every crawl stayed far from the ideal order. Out of window, PageRank on the latest allowances ranked later new approval pairs below the count of approving contracts in both windows, and for most spenders it is a reweighted count, as its closed form shows; AWP ranked later new senders slightly above the transfer in-degree overall and, in the first window, below it among spenders that a contract had already paid. The next steps are a discovery-order test on a later freeze, fixed before its data exist, importance labels from outside the chain, and a better detection of smart accounts.

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Supporting information

S1 Text. Event decoding, account types, balances, the discovery-order protocol, label formulas and extraction queries. Topic hashes and decoded fields of the ERC-20 events, the score windows and cohorts, the account-type rules with the smart-account rule and its limits, the balances and the balance cap, the token rule and the US-dollar valuation, graph weights and constants, the protocol of the discovery-order test in detail, exact formulas of every label, the same-window checks, the SQL patterns of the contract graph, and properties of the data that were not quantified.

S1 File. Excerpts of the analysis plan and of the W1 registration. The passages of the analysis plan (commit `d060f3a`) and of the W1 registration (commit `1747e64`, SHA-256 recorded in the extraction manifest) that this paper relies on, with every omitted passage marked, and the commit identifiers and SHA-256 hashes of the full files.

S1 Table. Data, graphs, cohorts and the four runs of the discovery-order test. (A) Events, account types, balances and the balance cap, the contract graph, the universes and hot sets, cohorts, labels and constants. (B) $\mathrm{AUC}_{10\%}$ of every ordering in the four runs. (C) The contrasts of items 1 and 2 of the protocol's list and the replication contrast in the three runs outside the decision. (D) Descriptive readings of the deciding run by ordering: hot contracts and summed importance reached, ties, wasted visits and solves.